

2023 Year 11 Yearly Exam - Mathematics Standard - Worked Solutions

1. **A**

From the reference sheet, the area of a sector is

$$A = \frac{\theta}{360} \pi r^2.$$

As the diameter is 12cm then the radius is 6 cm.

$$\text{So, } A = \frac{117}{360} \times \pi \times 6^2 \text{ cm}^2.$$

Hence, the correct option is **A**.

2. **C**

As the sum of the lowest and highest scores is 16 and the sum of the four scores is 30 this indicates that the sum of the 2 middle scores must be $30 - 16 = 14$.

Now, as there are 4 scores, the median will be the average of the middle two scores.

So, the median will be $14 \div 2 = 7$.

Hence, the correct option is **C**.

3. **D**

The absolute error is half of the smallest unit measured.

$$\text{The absolute error is } \frac{1}{2} \times 0.1 \text{ m} = 0.05 \text{ m}$$

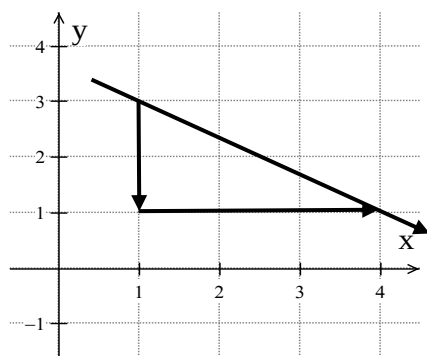
Therefore, the percentage error is

$$\pm \frac{0.05}{424.8} \times 100 = 0.01177024\ldots\%$$

$$= 0.0118\% \text{ (4 decimal places)}$$

Hence, the correct option is **D**.

4. **A**



In the diagram above we can see the y value decreasing by 2 and the x value increasing by 3, producing a gradient of $-\frac{2}{3}$. The only equation with a gradient of $-\frac{2}{3}$ is in option **A**.

Hence, the correct option is **A**.

5. **D**

For the boys the lower quartile is at 170 cm this means $\frac{3}{4}$ of the boys are over 170 cm. So, the number of boys at least 170 cm tall is $\frac{3}{4} \times 16 = 12$.

For the girls the median is at 170 this means $\frac{1}{2}$ of the girls are over 170 cm. So, the number of girls at least 170 cm tall is $\frac{1}{2} \times 14 = 7$.

Therefore, the number of students in the class least 170 cm tall is $12 + 7 = 19$.

Hence, the correct option is **D**.

6. **B**

The probability that the first apple selected is red is $\frac{5}{12}$.

If the first apple selected is red, then there are 4 red apples amongst the remaining 11 apples, so the probability of the second apple being red is $\frac{4}{11}$.

Therefore, the probability that Amanda selects 2 red apples is $\frac{5}{12} \times \frac{4}{11}$.

Hence, the correct option is **B**.

7. **A**

The line shown has a positive gradient with a rise of 7 and a run of 4, so its gradient is $+\frac{7}{4}$. It cuts the y axis at -7 , so in gradient intercept form, the equation would be $y = \frac{7}{4}x - 7$.

Hence, the correct option is **A**.

8. **C**

$$\begin{aligned} \text{Dividend yield} &= \frac{\text{dividend per share}}{\text{current price of one share}} \times 100 \\ &= \frac{1.20}{6.25} \times 100 = 19.2\% \end{aligned}$$

Hence, the correct option is **C**.

9. **D**

Each minute 4500 people leave the stadium this means after p minutes 4500 p people will leave the stadium.

This indicates that the number of people still in the stadium, after p minutes will be $N = 90\,000 - 4500p$.

Hence, the correct option is **D**.

10. **B**

The total increase over 6 years is

$$\$64\,000 - \$12\,000 = \$52\,000.$$

So, the average annual increase is

$$\$52\,000 \div 6 = \$8\,666.66 \dots$$

Therefore, the average percentage increase per year is

$$\frac{\$8\,666.66 \dots}{\$12\,000} \times 100 = 72.222 \dots \%$$

Hence, the correct option is **B**.

11. **C**

The volume of a sphere is $V = \frac{4}{3}\pi r^3$. This means the

volume of a hemisphere is $V = \frac{2}{3}\pi r^3$ and as the volume

of the hemisphere is $486\pi \text{ cm}^3$ then

$$\frac{2}{3}\pi r^3 = 486\pi \text{ that is } r^3 = 729 \text{ So, } r = 9 \text{ cm.}$$

The surface area of a sphere is $A = 4\pi r^2$ and this

indicates that the curved surface area of a

$$\text{hemisphere is } A = 2\pi r^2 = 2 \times \pi \times 9^2 = 162\pi \text{ cm}^2.$$

Hence, the correct option is **C**.

12. **B**

Applying the formula for Rachel to calculate the adult dosage A , we get

$$9.6 = \frac{8 \times A}{8 + 12} \text{ that is } 9.6 = \frac{8A}{20}$$

$$\text{So, } A = 9.6 \times 20 \div 8 = 24 \text{ mL.}$$

Now, the dosage D that Clara needs is

$$D = \frac{4 \times 24}{4 + 12} = \frac{96}{16} = 6 \text{ mL.}$$

Hence, the correct option is **B**.

13. **A**

The area of the whole rectangle is $2x \times 4x = 8x^2$

The diameter of the semicircle is $2x$ this means that its

radius is x . So, its area is $\frac{1}{2} \times \pi \times x^2$

Therefore, the shaded audience area is $8x^2 - \frac{1}{2}\pi x^2$.

Hence, the correct option is **A**.

14. **D**

Let the number of hours Kevin worked at time and a

half be x this means the number of hours he did at

normal time was $3x$. His total pay was \$774 this means

$$3x \times 21.5 + x \times 1.5 \times 21.5 = 774$$

$$64.5x + 32.25x = 774 \text{ that is } 96.75x = 774$$

$$\text{So, } x = 8.$$

Therefore, Kevin works 8 hours at time and a half and

24 hours at normal time which is a total of 32 hours.

Hence, the correct option is **D**.

15. **B**

The tax payable on a taxable income of \$110 000 is

$$\$5\,092 + 0.325 \times (\$110\,000 - \$45\,000) = \$26\,217.$$

The tax payable on a taxable income of \$135 000 is

$$\$29\,467 + 0.37 \times (\$135\,000 - \$120\,000) = \$35\,017.$$

Now, Layla will need to pay an extra tax of

$$\$35\,017 - \$26\,217 = \$8\,800$$

The increase of her income or the extra income

is \$25 000 this means the money she keeps from the

increase will be $\$25\,000 - \$8\,800 = \$16\,200$.

The percentage of the increase she will keep is

$$\frac{16\,200}{25\,000} \times 100 = 64.8\%$$

Hence, the correct option is **B**.

Question 16

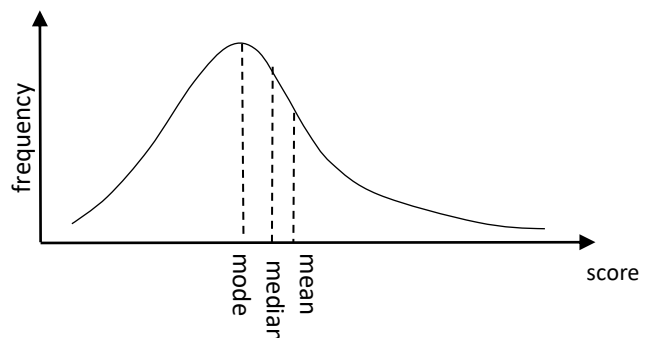
$$\text{a) } \bar{x} = \frac{\$250 + \$280 + \$305 + \$220 + \$420 + \$367}{6}$$

$$\text{So, } \bar{x} = \$307.$$

This could also have been found using the statistics mode on the calculator, and so no working would be necessary.

b) Using the statistics mode on the calculator, the standard deviation is 68.22 correct to 2 decimal places.

Question 17



A positively skewed distribution has a long tail at the higher scores end (right end) and the mean to the right of the median and the mode to the left of the median.

Question 18

As the TV depreciates by 20% every 6 months, it will depreciate 4 times in 2 years.

From the Reference Sheet,

$$S = V_0(1 - r)^n = \$2\,550(1 - 0.2)^4$$

$$\text{So, } S = \$1\,044.48.$$

Hence, the current value of this TV is \$1 044.48.

Question 19

The fraction of Gavin's employees who work at the main branch is $\frac{120}{320} = \frac{3}{8}$. This means the number of employees from the main branch who should be included in the sample should be $\frac{3}{8} \times 40 = 15$.

Question 20

The absolute error is half of the smallest unit measured.

So, the absolute error is $\frac{1}{2} \times 1 \text{ cm} = 0.5 \text{ cm}$

The lower limit of the area of the rectangular piece of wood is $47.5 \times 23.5 = 1\,116.25 \text{ cm}^2$ and the upper limit of its area is $48.5 \times 24.5 = 1\,188.25 \text{ cm}^2$.

Question 21

Joanne's fortnightly wage is \$2 460. So, in 4 weeks she will earn $\$2\,460 \times 2 = \$4\,920$.

Now, her holiday loading is 17.5% of four weeks' pay this means her holiday loading is

$$0.175 \times \$4\,920 = \$861$$

Hence, her total holiday pay for four weeks is

$$\$4\,920 + \$861 = \$5\,781$$

Question 22

Jared will iron his clothes $40 \div 5 = 8$ times in 40 days.

This means the total time he spent ironing his clothes will be $8 \times 36 = 288 \text{ minutes} = 4.8 \text{ hours}$

Now, the electricity the iron will use

$$2.7 \text{ kW} \times 4.8 \text{ hours} = 12.96 \text{ kWh}$$

Hence, the cost of his ironing is

$$12.96 \times 27.5 \text{ cents} = \$3.56.$$

Question 23

Kate will receive commission on

$$\$12\,100 - \$3\,200 = \$8\,900.$$

Her commission in this week will be

$$0.085 \times \$8\,900 = \$756.50$$

This week her wage will be

$$\$756.50 + \$420 = \$1\,176.50.$$

Question 24

As the investment is compounded monthly this means the interest rate r should be expressed per month that is

$$r = 3\% \text{ p. a.} = \frac{3}{12}\% = 0.0025 \text{ per month}$$

Also, the number of compounding period n should be the number of months that is $n = 4 \times 12 = 48 \text{ months}$.

From the Reference Sheet, $FV = PV(1 + r)^n$

$$FV = \$80\,000 (1 + 0.0025)^{48}$$

So, $FV = \$90\,186.24$.

Hence, the total amount of interest Ricardo earns is

$$\$90\,186.24 - \$80\,000 = \$10\,186.24$$

Question 25

First, we need to find the lower and upper quartile to the interquartile range.

Lower Quartile

Upper Quartile

$$Q_1 = \frac{30 + 30}{2} = 30$$

$$Q_3 = \frac{42 + 42}{2} = 42$$

So, the interquartile range is $42 - 30 = 12$.

Therefore, the highest score before having an outlier is

$$Q_3 + 1.5 \times IQR = 42 + 1.5 \times 12 = 60.$$

Hence, 64 is an outlier for this set of scores.

Question 26

City B is $+4 - (-10) = 14$ hours ahead of City A.

When the plane left City A it was Friday at 4:20 AM, which means the time in City B was 18:20 Friday as City B is 14 hours ahead of City A.

The flight arrived in City B 12 hours after 18:20 Friday.

Now, 6 hours after 18:20 Friday it would be 00:20

Saturday and 6 hours after this time it would be

6:20 am on Saturday.

Hence, Jane's plane will arrive at 6:20 am on Saturday.

Question 27

a) The salmon has 220 kJ in 55g this means it contains

$220 \div 55 = 4 \text{ kJ}$ in 1 g. Ronald ate 176 g of salmon this means the energy he took from salmon was

$$4 \times 176 = 704 \text{ kJ}.$$

Now, the pasta has 1 020 kJ in 120 g this means it

contains $1\,020 \div 120 = 8.5 \text{ kJ}$ in 1 g. So, the energy he took from the pasta was $48 \times 8.5 = 408 \text{ kJ}$

Hence, the total number of kilojoules in Ronald's food was $704 + 408 = 1\,112 \text{ kJ}$.

b) Since 1 kilocalorie = 4.184 kJ then

$$1\,112\text{ kJ} = \frac{1\,112}{4.184} = 265.77 \text{ kilocalories.}$$

As Ronald weighs 74 kg he will burn 8.1 kilocalories per minute when he plays tennis.

Hence, he will need to play tennis for

$265.77 \div 8.1 \approx 33$ minutes to use all the energy in his in the salmon and the pasta.

Question 28

a) The time the boat was hired is

$4 \text{ hours } 15 \text{ minutes} = 4.25 \text{ hours.}$

Hence, the cost to hire a speed boat from Boat champions is

$$C = 65.6 \times 4.25 + 98.4 = \$377.20$$

b) Let time that Tania hired the boat be T .

Now, as the difference between the costs of hiring the boat from the two companies is \$13.70 this means

Cost Boat Kingdom – Cost Boat Champions = \$13.70

that is $72.3T + 78.6 - (65.6T + 98.4) = 13.70$

$$6.7T - 19.8 = 13.70 \text{ So, } T = 5 \text{ hours}$$

Hence, Tania saved \$13.70 by hiring a speed boat for 5 hours from Boat Champions instead of hiring it from Boat Kingdom.

Question 29

Jennifer's deposit was $40\% \times \$32\,000 = \$12\,800$.

This means the remaining amount she still owing was

$$\$32\,000 - \$12\,800 = \$19\,200$$

As she was given 6 months interest free this indicates that she needed to pay interest only for 1 year 6 months or 1.5 years instead of two years.

Now, the simple interest charged on the amount still owing was $I = \$19\,200 \times 0.04 \times 1.5 = \$1\,152$.

Therefore, the total amount she needed to pay was

$$\$19\,200 + \$1\,152 = \$20\,352.$$

The money is to be repaid over 2 years which is 24 months.

Hence, Jennifer's monthly repayment must be

$$\$20\,352 \div 24 = \$848.$$

Question 30

As the intensity I varies inversely with the resistance R

then $I = \frac{k}{R}$, where k is a constant.

Substituting $I = 24$ when $R = 6$, we get

$$24 = \frac{k}{6} \text{ that is } k = 144. \text{ Therefore } I = \frac{144}{R}.$$

$$\text{Now, when } R = 8, I = \frac{144}{8} = 18.$$

Hence, $I = 18$.

Question 31

a) The area of rectangle ABCD is

$$16 \times 12 = 192 \text{ cm}^2$$

The area of the semicircle is

$$\frac{1}{2} \times \pi \times 4^2 = 8\pi \text{ cm}^2$$

Hence, the area of the cross section is

$$A = 192 - 8\pi = 166.8673 \text{ cm}^2$$

b) The volume of the prism is

$$V = 166.8673 \times 16 = 2669.8768 \text{ cm}^3.$$

Question 32

As Matthew's allowable deductions are 9% of his gross income then his taxable income must be 91% of his gross income this means his taxable income is $\$175\,000 \times 0.91 = \$159\,250$.

Hence, Matthew's tax payable on his taxable income is

$$\begin{aligned} \text{Tax} &= (\$159\,250 - \$120\,000) \times 0.37 + \$29\,467 \\ &= \$14\,522.50 + \$29\,467 \\ &= \$43\,989.50. \end{aligned}$$

Question 33

The table below show all the outcomes when a pair of dice are rolled once.

The shaded cells in the table shows when the sum of the numbers on the uppermost faces is at least 10.

	1	2	3	4	5	6
1						
2						
3						
4						
5						
6						

This indicates the probability to get a sum of at least 10 in one roll is $P(\text{sum} \geq 10) = \frac{6}{36} = \frac{1}{6}$.

So, the probability to a get a sum at least 10 on both rolls is $P(\text{both rolls} \geq 10) = \frac{1}{6} \times \frac{1}{6} = \frac{1}{36}$.

Hence, the probability that Patricia wins a major prize is $\frac{1}{36}$.

Question 34

Malcolm stopped drinking at 11:20 pm and left the party 2:50 am when BAC reached zero. This means he needed to wait 3.5 hours after his last drink to reach a BAC of zero. By substituting 3.5 in

$$\text{Number of hours for BAC to reach zero} = \frac{\text{BAC}}{0.015},$$

$$\text{we get } 3.5 = \frac{\text{BAC}}{0.015} \text{ this means}$$

$$\text{BAC} = 3.5 \times 0.015 = 0.0525.$$

Also, the number of standard drinks that Malcolm consumed is $N = 3 \times 1.5 = 4.5$.

Now, by substituting $\text{BAC}_{\text{Male}} = 0.0525$, $N = 4.5$ and Malcolm's mass $M = 78 \text{ kg}$ into

$$\text{BAC}_{\text{Male}} = \frac{10N - 7.5H}{6.8M}, \text{ we get}$$

$$0.0525 = \frac{10 \times 4.5 - 7.5H}{6.8 \times 78} \text{ that is } 0.0525 = \frac{45 - 7.5H}{530.4}$$

By multiplying by 530.4 on both sides, we get

$$27.846 = 45 - 7.5H \text{ this means } -17.154 = -7.5H$$

So, $H = 2.2872 \text{ hours} \approx 2 \text{ hours } 17 \text{ minutes}$.

As Malcolm stopped drinking at 11:20 pm then he started to drink 2 hours 17 minutes before this time.

Now, 2 hours before 11:20 pm is 9:20 pm and 17 minutes before 9:20 pm is 9:03 pm.

Hence, Malcolm started his first drink at 9:03 pm.

Question 35

Area of the first trapezium

$$A_1 = \frac{100}{2}(h + 6 + h + 10)$$

$$A_1 = 50(2h + 16). \text{ Therefore, } A_1 = 100h + 800$$

Area of the second trapezium

$$A_2 = \frac{100}{2}(h + 10 + h + 8)$$

$$A_2 = 50(2h + 18). \text{ Therefore, } A_2 = 100h + 900$$

So, the total area of Kevin's land is

$$A = 100h + 800 + 100h + 900$$

$$A = 200h + 1700$$

As the total area of Kevin's land is $6\,100 \text{ m}^2$ then

$$6\,100 = 200h + 1700 \text{ that is } 4\,400 = 200h$$

$$\text{Hence, } h = 22 \text{ m.}$$

Question 36

At first the ratio of black cars to white cars was 5 : 2 this means if we let the number of black cars at first be $5n$ then the number of white cars at that time should be $2n$.

After the changes, the number of black cars became $5n - 10$ and the number of white cars became $2n + 14$.

Using the new ratio of black cars to white cars, we get

$$\frac{5n-10}{2n+14} = \frac{8}{5} \text{ this indicates } 25n - 50 = 16n + 112$$

$$\text{that is } 9n = 162. \text{ Therefore, } n = 18.$$

Now, as the number of black cars in the car dealership at first was $5n$ then this number was $5 \times 18 = 90$ black cars.